UML DIAGRAMS

Introduction to UML

UML (Unified Modeling Language) is a general-purpose, graphical modeling language in the field of Software Engineering. UML is used to specify, visualize, construct, and document the artifacts (major elements) of the software system. It was initially developed by James Rumbaugh in 1994-95 at Rational software, and its further development was carried out through 1996. In 1997, it got adopted as a standard by the Object Management Group.

What is UML

The UML stands for Unified modelling language, is a standardized general-purpose visual modelling language in the field of Software Engineering. It is used for specifying, visualizing, constructing, and documenting the primary artifacts of the software system. It helps in designing and characterizing, especially those software systems that incorporate the concept of Object orientation. It describes the working of both the software and hardware systems.

Goals of UML

- Since it is a general-purpose modelling language, it can be utilized by all the modelers.
- UML came into existence after the introduction of object-oriented concepts to systemize and consolidate the object-oriented development, due to the absence of standard methods at that time.
- The UML diagrams are made for business users, developers, ordinary people, or anyone who is looking forward to understand the system, such that the system can be software or non-software.
- Thus it can be concluded that the UML is a simple modelling approach that is used to model all the practical systems.

Characteristics of UML

The UML has the following features:

- It is a generalized modelling language.
- It is distinct from other programming languages like C++, Python, etc.
- It is interrelated to object-oriented analysis and design.
- It is used to visualize the workflow of the system.
- It is a pictorial language, used to generate powerful modelling artifacts.

Diagrams

The diagrams are the graphical implementation of the models that incorporate symbols and text. Each symbol has a different meaning in the context of the UML diagram. There are thirteen different types of UML diagrams that are available in UML 2.0, such that each diagram has its own set of a symbol. And each diagram manifests a different dimension, perspective, and view of the system.

UML diagrams are classified into three categories that are given below:

1. Structural Diagram
2. Behavioural Diagram
3. Interaction Diagram

Structural Diagram: It represents the static view of a system by portraying the structure of a system. It shows several objects residing in the system. Following are the structural diagrams given below:

1. *Class diagram*

Software engineers use class diagrams to show the classes, attributes, methods in a system and to describe their relationships to each other. Class diagrams enable developers to sketch a static view of a system before creating it.

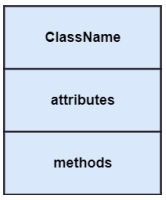
Purpose of Class Diagrams

The main purpose of class diagrams is to build a static view of an application. It is the only diagram that is widely used for construction, and it can be mapped with object-oriented languages. It is one of the most popular UML diagrams. Following are the purpose of class diagrams given below:

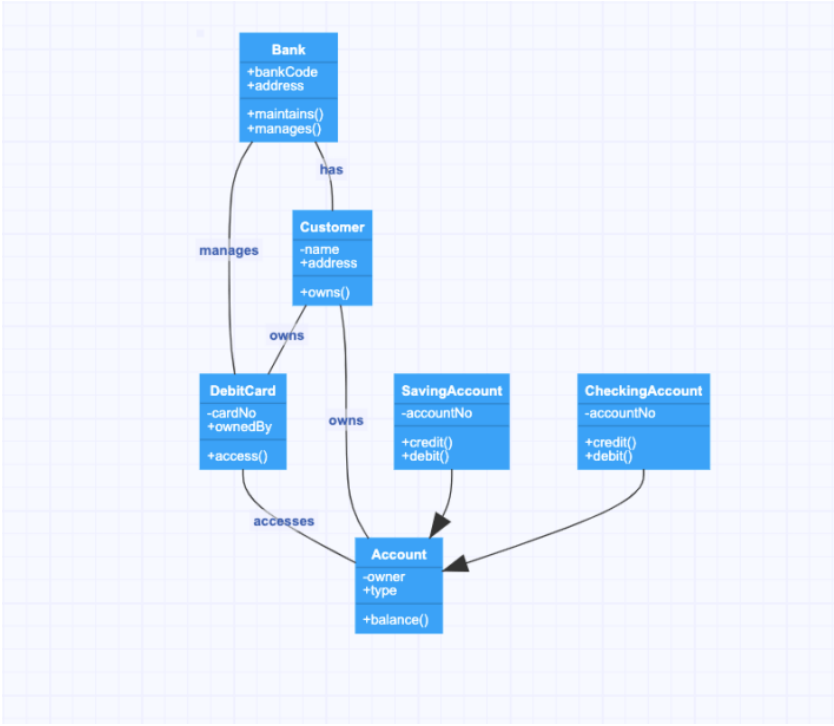
1. It analyses and designs a static view of an application.
2. It describes the major responsibilities of a system.

3. It is a base for component and deployment diagrams.
4. It incorporates forward and reverse engineering.

Notation of an Class Diagram



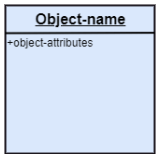
Example: Class Diagram of Bank Application



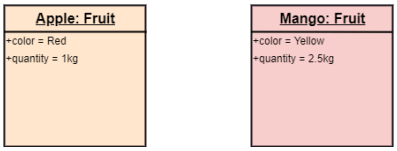
2. Object diagram

Object diagrams are dependent on the class diagram as they are derived from the class diagram. It represents an instance of a class diagram. The objects help in portraying a static view of an object-oriented system at a specific instant.

Notation of an Object Diagram



Example of Object Diagram

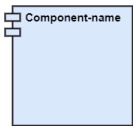


3. Component diagram

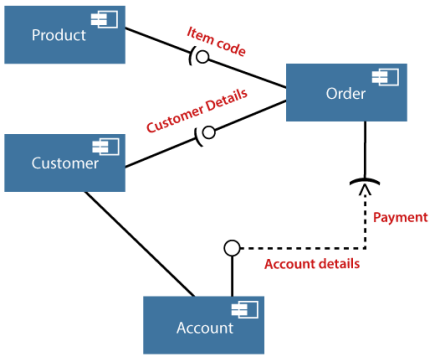
A component diagram is used to break down a large object-oriented system into the smaller components, so as to make them more manageable. It models the physical view of a system such as executables, files, libraries, etc. that resides within the node.

Notation of a Component Diagram

a) A component



Example of Component Diagram



4. Deployment diagram

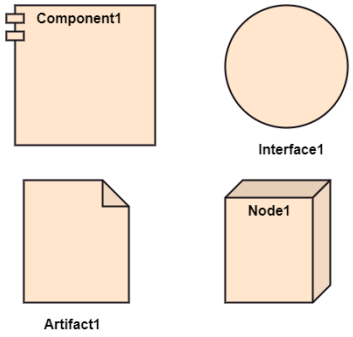
The deployment diagram visualizes the physical hardware on which the software will be deployed. It portrays the static deployment view of a system. It involves the nodes and their relationships.

It ascertains how software is deployed on the hardware. It maps the software architecture created in design to the physical system architecture, where the software will be executed as a node. Since it involves many nodes, the relationship is shown by utilizing communication paths.

Symbol and notation of Deployment diagram

The deployment diagram consists of the following notations:

1. A component
2. An artifact
3. An interface
4. A node



How to draw a Deployment Diagram?

The deployment diagram portrays the deployment view of the system. It helps in visualizing the topological view of a system. It incorporates nodes, which are physical hardware. The nodes are used to execute the artifacts. The instances of artifacts can be deployed on the instances of nodes.

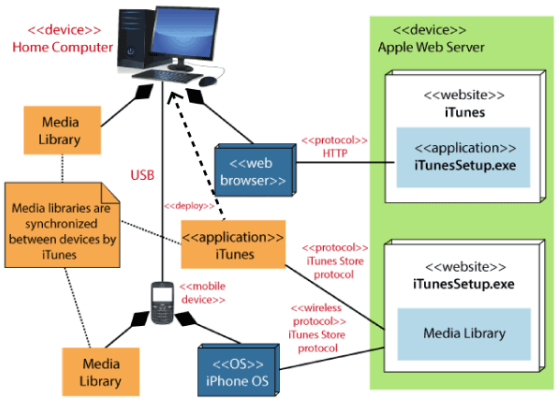
Since it plays a critical role during the administrative process, it involves the following parameters:

1. High performance
2. Scalability
3. Maintainability
4. Portability
5. Easily understandable

One of the essential elements of the deployment diagram is the nodes and artifacts. So it is necessary to identify all of the nodes and the relationship between them. It becomes easier to develop a deployment diagram if all of the nodes, artifacts, and their relationship is already known.

Example of a Deployment diagram

A deployment diagram for the Apple iTunes application is given below.



Behavioral Diagram: It depicts the behavioral features of a system. It deals with dynamic parts of the system. It encompasses the following diagrams:

1. Activity diagram

In UML, the activity diagram is used to demonstrate the flow of control within the system rather than the implementation. It models the concurrent and sequential activities.

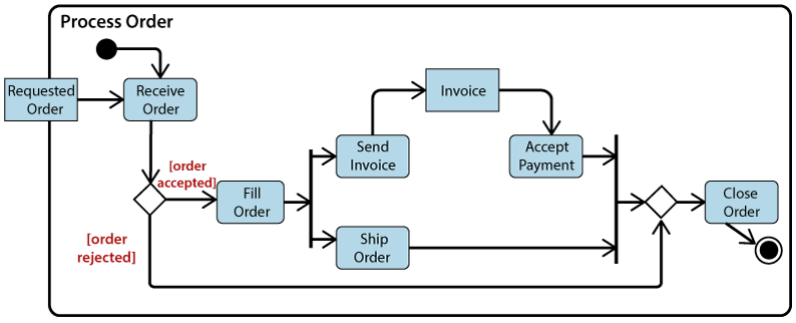
The activity diagram helps in envisioning the workflow from one activity to another. It put emphasis on the condition of flow and the order in which it occurs. The flow can be sequential, branched, or concurrent, and to deal with such kinds of flows, the activity diagram has come up with a fork, join, etc.

It is also termed as an object-oriented flowchart. It encompasses activities composed of a set of actions or operations that are applied to model the behavioral diagram.

Example of an Activity Diagram

An example of an activity diagram showing the business flow activity of order processing is given below.

Here the input parameter is the Requested order, and once the order is accepted, all of the required information is then filled, payment is also accepted, and then the order is shipped. It permits order shipment before an invoice is sent or payment is completed.



When to use an Activity Diagram?

An activity diagram can be used to portray business processes and workflows. Also, it used for modeling business as well as the software. An activity diagram is utilized for the followings:

1. To graphically model the workflow in an easier and understandable way.
2. To model the execution flow among several activities.
3. To model comprehensive information of a function or an algorithm employed within the system.
4. To model the business process and its workflow.
5. To envision the dynamic aspect of a system.
6. To generate the top-level flowcharts for representing the workflow of an application.
7. To represent a high-level view of a distributed or an object-oriented system.

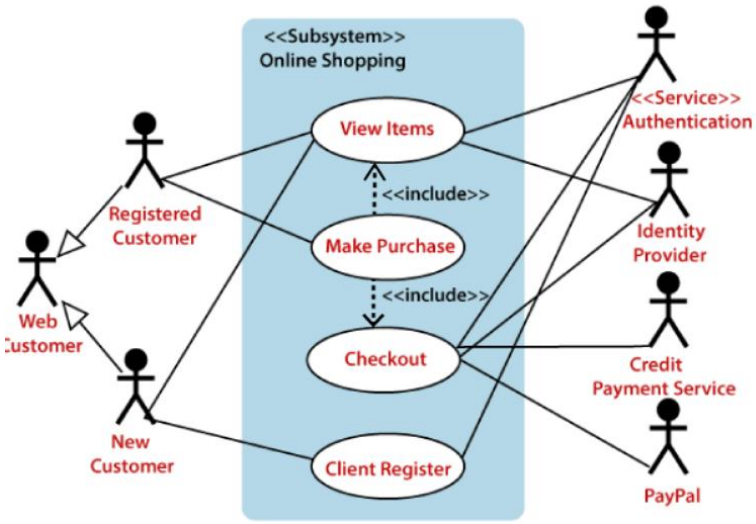
2. Use case diagram

A use case diagram is used to represent the dynamic behaviour of a system. It encapsulates the system's functionality by incorporating use cases, actors, and their relationships. It models the tasks, services, and functions required by a system/subsystem of an application. It depicts the high-level functionality of a system and also tells how the user handles a system.

Example of a Use Case Diagram

A use case diagram depicting the Online Shopping website is given below.

Here the Web Customer actor makes use of any online shopping website to purchase online. The top-level uses are as follows; View Items, Make Purchase, Checkout, Client Register. The **View Items** use case is utilized by the customer who searches and view products. The **Client Register** use case allows the customer to register itself with the website for availing gift vouchers, coupons, or getting a private sale invitation. It is to be noted that the **Checkout** is an included use case, which is part of **Making Purchase,** and it is not available by itself.



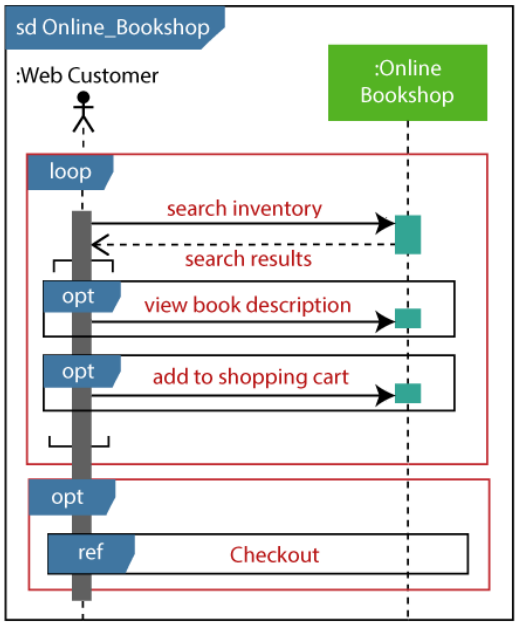
Interaction diagram: It is a subset of behavioural diagrams. It depicts the interaction between two objects and the data flow between them. Following are the several interaction diagrams in UML:

1. Sequence diagram

   The sequence diagram represents the flow of messages in the system and is also termed as an event diagram. It helps in envisioning several dynamic scenarios. It portrays the communication between any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time. In UML, the lifeline is represented by a vertical bar, whereas the message flow is represented by a vertical dotted line that extends across the bottom of the page. It incorporates the iterations as well as branching.

Example of a Sequence Diagram

An example of a high-level sequence diagram for online bookshop is given below. Any online customer can search for a book catalog, view a description of a particular book, add a book to its shopping cart, and do checkout.



2. Collaboration diagram

The collaboration diagram is used to show the relationship between the objects in a system. Both the sequence and the collaboration diagrams represent the same information but differently. Instead of showing the flow of messages, it depicts the architecture of the object residing in the system as it is based on object-oriented programming. An object consists of several features. Multiple objects present in the system are connected to each other. The collaboration diagram, which is also known as a communication diagram, is used to portray the object's architecture in the system.

Components of a collaboration diagram

